

Reducing Crime Without the Police? Evidence from a Place-Based Infrastructure Intervention in Bogotá

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The policing trap in the developing world

Crime imposes enormous welfare costs ([Jaitman et al. 2017](#)) — yet policing often fails in the developing world

- Prevalence of organized crime: lucrative illicit markets produce corruption and heavily armed groups
- Patrol misallocation; hotspot effects mixed ([Mastrobuoni 2020](#))
- Corruption undermines trust ([Blair et al. 2021](#))
- Militarized interventions aren't effective and produce abuses ([Blair and Weintraub 2023](#), [Magaloni et al. 2022](#))
- Understaffed: Bogotá has 65% of the UN-recommended police per capita

One alternative: transform the **built environment**.

Can changes to the built environment improve public safety?

- **Small-scale** interventions show promise: street lighting, tree cover, securing abandoned buildings (Branas et al. 2016, Chalfin et al. 2022)
- But evidence on **large-scale transport infrastructure** is limited (Montolio 2018, Phillips and Sandler 2015)
- And crime effects are rarely incorporated in the **welfare evaluation** of transport projects: travel-time savings and ridership dominate

Meanwhile, the public already has a strong prior...

“Transit brings crime”: an old fear

Atlanta debates MARTA rail expansion, 1997

"Having MARTA [rail] increases the chances of crime. The criminals can get off the train and break into your home and get back onto the train. I don't think that's as likely to happen on buses, where they have to walk past the driver to get on board."

Source: Atlanta Journal Constitution (1997)

The “crime train”: alive and well today

- The trope is over a century old (Bloomberg 2023)
- Communities resist transit expansions fearing they *import* crime
- And the fear has a rational core: transit lowers travel costs for offenders too



Source: Bloomberg (2023)

Research question: Can large-scale **urban infrastructure** reduce crime *without* the police — or does it spread crime through the network?

TransMiCable: a gondola lift on Bogotá's hillside

The system

- Gondola connecting Ciudad Bolívar to the TransMilenio (world's largest BRT)
- 3.4 km, **3 hillside stations** + Tunal base terminal
- Construction 2016–2018; inaugurated December 2018 at cost USD 109 million
- Commute time from 60 min → 15–20 min



The line over the Ciudad Bolívar hillside



Juan Pablo II station

This paper

What we do

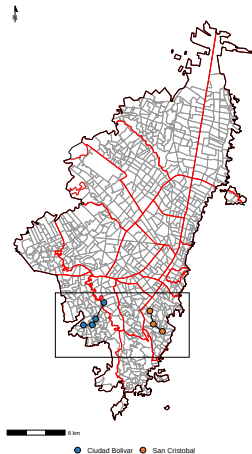
- **Design:** spatial difference-in-differences at the 100 m grid-cell level — cells ≤ 800 m of a station vs. cells 800 m–2 km, before and after construction and opening
- **Data:** geocoded incident-level police records (2011–2020), FBI UCR property and violent categories

What we find

1. Total crime falls $\approx$ **16%** near stations (property —14%, violent —7%)
2. Decline begins **during construction** and tracks **operating hours**: informal monitoring, not commuting flows
3. Not physical upgrading: satellite building data show **no differential construction or densification** near stations
4. Displacement downstream is real but small

Why Ciudad Bolívar?

- More than 700000 residents; among Bogotá's poorest localities
- Homicide rate of 42 per 100k (1.7× national avg.)
- Steep Andean terrain: historically transit-isolated
- Jobs concentrated in the city center and north; unequal transit access compounds the gap
- Promise of the gondola: mobility, state presence, and — possibly — safety



Why might transit infrastructure reduce crime?

What offenders choose

A potential offender weighs three options: commit a crime **near the station**, commit a crime **further along the transit network**, or pursue **legal work**.

The choice depends on expected returns, the risk of getting caught, and the cost of reaching the most lucrative targets.

Net effect **ambiguous a priori**: deterrence near stations vs. easier access downstream.
The model predicts *where* and *when* each dominates.

What TransMiCable changes

Near the station

More eyes on the street → higher detection risk

Better job access → higher value of legal work locally

Downstream stops

Cheaper and faster to reach → more attractive

More anonymous than the station → lower detection risk

But outside work options downstream also more accessible

Three predictions

P1 — Local deterrence

More monitoring + better
job access decreases
crime near stations

P2 — Early onset

Construction itself raises
surveillance around sites
When: effects begin even
before service starts

P3 — Displacement

Cheaper access to the
network means crime
increases at downstream
stops

When: only post-opening

Data and spatial difference-in-differences

Data

- Geocoded admin crime records, Colombian Police (2011–2020)
- FBI UCR: 5 property + 3 violent sub-categories
- 100 m grid cells: 9,252 cells across 32 quarters, with 77 neighborhood clusters

Treatment vs. control

- Treated: cells ≤ 800 m of a hillside station
- Control: cells 800 m–2 km from any station
- Staggered construction (Q3/Q4 2016, Q1 2017)

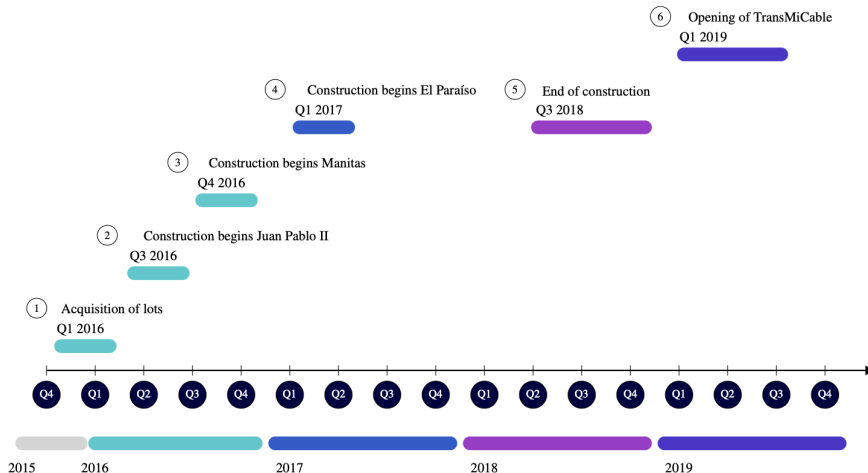
Baseline DiD (post-opening)

$$\text{Crime}_{g,t} = \beta \cdot \mathbf{1}[d_g \leq 800] \times \text{Post}_t + \alpha_g + \delta_t + \varepsilon_{g,t}$$

Identification

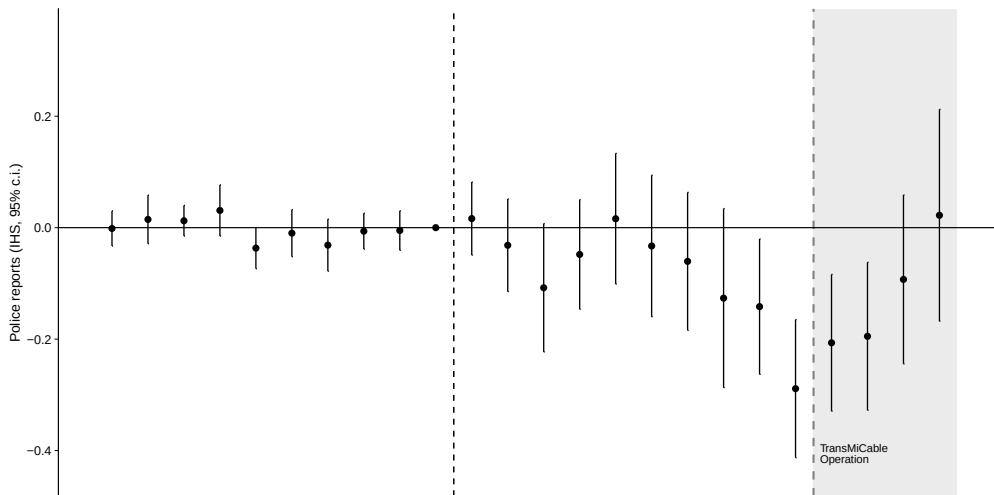
- Flat pre-trends in the event study
- Triple-DiD: San Cristóbal had *planned but unbuilt* stations — same geography, same time period, no treatment. Zero effect there rules out common shocks to peripheral Bogotá.
- Gardner (2022) staggered estimator avoids negative-weight bias (Goodman-Bacon 2021)

Staggered construction, one opening date



Crime falls, and begins to fall *before* the gondola opens

Gardner (2022) staggered event study, treated ≤ 800 m vs. control 800 m–2 km



Decline begins around construction start dates — before any passenger boards.

Construction-phase deterrence!

TransMiCable reduces crime: baseline estimates

Outcome	$\hat{\beta}$	(s.e.)
Total crime	−0.165***	(0.048)
Property crime	−0.137**	(0.062)
Violent crime	−0.072**	(0.029)

IHS of quarterly crime counts; grid-cell and quarter-year FE; SEs clustered at neighborhood.

- Total crime falls $\approx$ **16%** in the station catchment post-opening
- Driven by **property crime** (−14%); violent crime −7%
- Robust across estimators: OLS, Poisson, negative binomial [► Estimators](#)
- No effect in San Cristóbal, where stations were planned but *not built* (triple-DiD): rules out common shocks to peripheral Bogotá

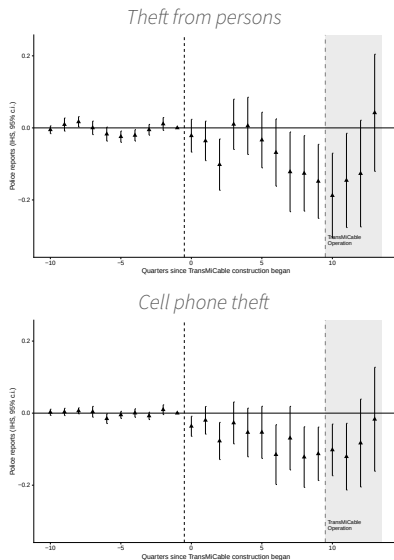
The decline is two offenses

Theft from persons & cell phone theft — Gardner (2022) ATTs by UCR sub-category

	Construction	Post-opening
Theft from persons	−0.049***	−0.151***
Cell phone theft	−0.063***	−0.101***
Motor vehicle theft	−0.001	+0.005**
Motorcycle theft	+0.006**	+0.003
Burglary	−0.003	+0.011*
Personal injuries	−0.006	−0.036**
Domestic violence	−0.003	−0.047***
Sexual offenses	+0.001	−0.003

Only offenses committed in **contested public space** respond — and they respond already during construction.

Private targets (vehicles at rest, homes): flat throughout.



Crime reduction tracks ridership patterns

Gardner (2022) construction-aligned event study, weekdays only

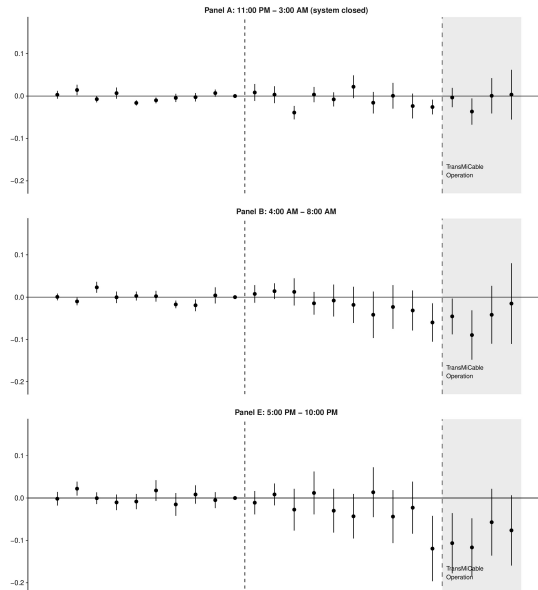
Mechanism test

If informal monitoring drives the decline, effects must track hours when the station is active — and disappear when it closes.

Evidence

- **Panel A** (11 pm–3 am, *closed*): no effect
- **Panel B** (4–8 am, *morning rush*): strong decline
- **Panel E** (5–10 pm, *evening peak*): strong decline

Hard to reconcile with neighborhood investment or income effects alone.



Not neighborhood upgrading: no building boom near stations

Satellite building footprints, Google Open Buildings Temporal (2016–2023)

Alternative story

Crime fell because the gondola triggered redevelopment — new construction, densification, gentrification — not monitoring.

Test

- Annual building footprints and heights, 4 m resolution satellite data
- Same DiD design as crime analysis: catchment (≤ 800 m) vs. ring (800 m–2 km), cell + year FE

The crime decline is not explained by physical redevelopment. Consistent with monitoring: effects track operating hours, not fixed upgrades.

Treated $\times$	$\ln(\text{bldg. count})$	$\text{asinh}(\text{height})$
Construction	−0.015	+0.008
Post-opening	−0.006	−0.017

Point estimates $\approx$ zero, if anything slightly *negative*.

No differential construction or densification in the catchment — during construction or after opening.

Where could crime displace? Two margins downstream

The model predicts: cheaper network access moves crime **downstream** — and only **post-opening**. Two natural margins:

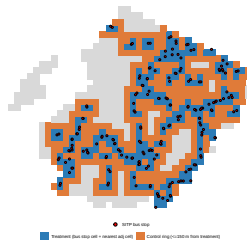
1. Tunal terminal



Portal Tunal: the gondola's only connection to the BRT network

Concentrates transfers, crowds, waiting time → 200 m distance bands around the node

2. SITP feeder bus stops



Blue: treated (stop cell + nearest). Orange: control (≤ 150 m).

Last-mile stops, thin surveillance → narrow bands (stops 200–300 m apart); timing inherited from nearest station

At Tunal, theft concentrates at the node — violence falls

Gardner (2022), 200 m distance bands around the terminal, post-opening

Distance to Tunal	Violent	Property
≤ 200 m	-0.330***	+0.307*
200–400 m	-0.149***	-0.125
400–600 m	-0.312***	+0.081
600–800 m	-0.011	+0.106

Property crime ≤ 200 m is elevated already during construction (+0.291***).

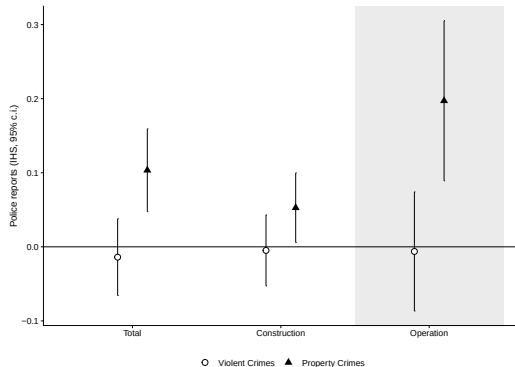
- Property crime rises only in the **immediate micro-environment**: crowding, transfers, short-duration interactions
- Violent crime **falls across all bands**: monitoring extends around the node
- Non-monotonic pattern = displacement *toward the node*, not a uniform spillover

Displacement to feeder stops: offense-specific and small

Feeder stops: property $\uparrow$ once operations begin; violent crime flat in both phases

Internal consistency

- Same two categories that fell in the station catchment rise at feeder stops
- Motor vehicle, motorcycle, burglary: flat everywhere
- Domestic violence at feeder stops: zero — the placebo passes



Property $\uparrow$ at feeder stops post-operation; violent crime flat.

Displacement offsets only $\approx 6\%$ of catchment gains.

The “crime train” is real — but small and predictable.

Corridor accounting: $\approx 13,000$ fewer property crimes

Margin	Cells	$\hat{\beta}$ per cell, per quarter	Total Δ crimes, Q1–Q4 2019
Catchment (3 hillside stations)	6,898	-0.50^{***}	$-13,842$
Tunal terminal, 0–200 m	39	$+1.21^*$	$+189$
Tunal terminal, 200–800 m (net)	1,422	—	$+415$
Feeder bus stops (SITP)	213	$+0.25^{**}$	$+213$
Net corridor effect			$-13,026$

The **network** is the right welfare unit.

Catchment-only evaluations overstate benefits. Displacement is predictable: good news for public policy.

Conclusion

Findings

1. Crime falls $\approx$ **16%** near stations, driven by public-space thefts
2. Begins **during construction** and tracks **operating hours**: staffed-station monitoring
3. No building boom (not gentrification);
4. Displacement offense-specific and small

Implications

Infrastructure **can** reduce crime without the police.

And the “crime train”? Real but modest, predictable, and far outweighed by the benefits.



Thank you!

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Appendix

Why not standard TWFE?

Staggered adoption requires a heterogeneity-robust estimator

The problem with TWFE

Staggered treatment + heterogeneous effects $\Rightarrow$ long-treated units receive *negative weights* when used as controls for newly treated ones.

Estimates can be biased *in sign*, not just magnitude.

Our solution: Gardner (2022)

1. Fit unit + time FE on *never-treated* cells only.
2. Subtract predicted counterfactual $\Rightarrow$ residualised panel.
3. Recover cohort-by-event-time effects

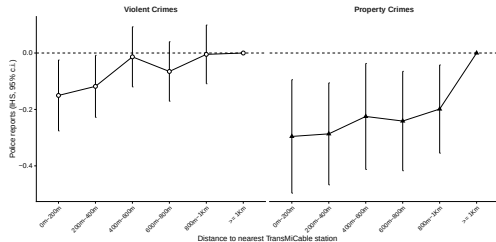
	TWFE	Gardner
Hetero. robust	×	✓
Negative weights	possible	none
Treatment reversal	problem	✓
Pre-trend test	✓	✓

Pre-treatment estimates ≈ 0 under both estimators.

Also robust to treatment reversal — relevant for the feeder-stop analysis.

The 800 m catchment: spatial decay by crime type

- Violent crime: concentrated within **400 m**, then flat
- Property crime: gradient persists to **800 m+**
- 800 m = avg. walk to transit in Bogotá (Muñoz-Raskin 2010)
- Cutoff **conservative**: control cells bear residual treatment $\Rightarrow$ property estimates biased toward zero

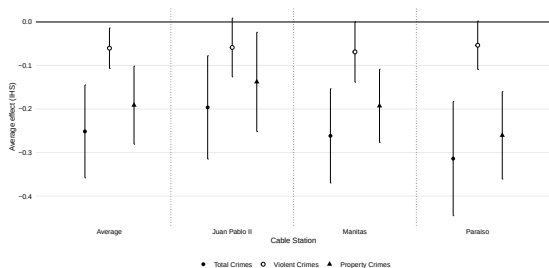


Violence requires direct contact $\rightarrow$ responds only to localised monitoring.
Opportunistic theft responds to a broader pedestrian footprint.

No heterogeneity across stations

Callaway & Sant'Anna (2021) group-time ATTs by station

- Juan Pablo II, Manitas, Mirador del Paraíso all show consistent crime declines
- Effect not idiosyncratic to any single site



Callaway-Sant'Anna ATTs (IHS)

Station	Total	Property
Average	-0.26	-0.21
Juan Pablo II	-0.23	-0.18
Manitas	-0.25	-0.21
Mirador	-0.29	-0.24

Robustness: results hold across estimators

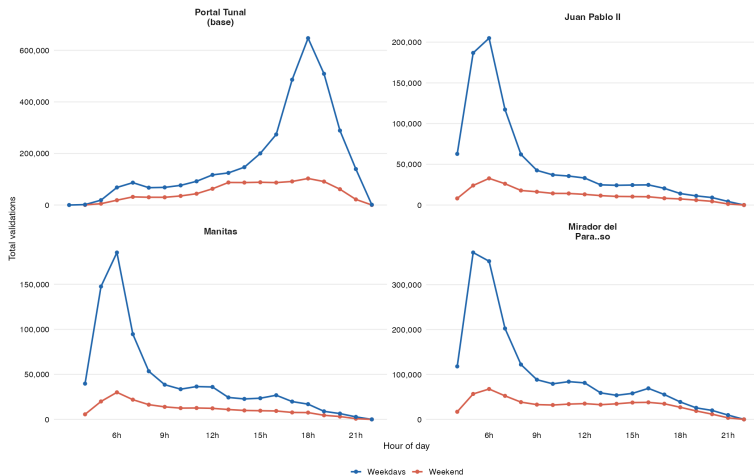
	IHS (main)	OLS	Poisson	Neg. Bin.
Total crime	−0.165***	−0.470***	−0.296***	−0.304***
Violent crime	−0.072**	−0.097*	−0.191**	−0.257***
Property crime	−0.137**	−0.373**	−0.339***	−0.339***

Grid-cell and quarter-year FE in all columns; SEs clustered at neighbourhood. Signs and significance consistent across all functional forms.

[◀ Back to baseline results](#)

Ridership: validations by hour of day, by station

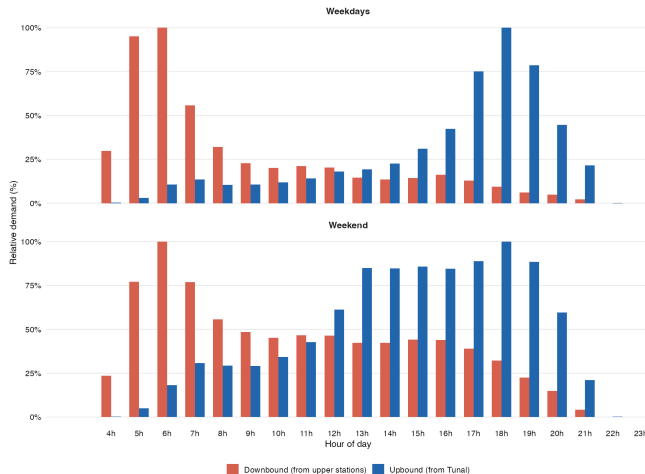
Fare validation records, Transmilenio S.A.



Hillside stations peak at the **morning rush** (5–6 am, residents descend to work); Tunal peaks at the **evening return** (~6 pm). Predictable, concentrated pedestrian flows.

Ridership: peak-hour profiles by direction of travel

Demand normalized to each direction's peak hour (100% = peak)



Weekday commuter asymmetry: downbound peaks 5–6 am, upbound peaks 6 pm — the flows behind the time-of-day heterogeneity in the mechanism test.